

Statistical and Machine Learning

Neural Networks for Tabular Data

Instructions

All questions should first be discussed in pairs or small groups. Afterwards, we shall discuss them together. Solutions will be distributed after each session.

Question 1

Many simple models, such as linear and logistic regression, are linear in the input $\mathbf{x}$. One way to make them more flexible is **basis function expansion**, replacing $\mathbf{x}$ with $\phi(\mathbf{x})$ before the linear step.

- Give an example of such a transform ϕ for a one-dimensional input.
- What is the main limitation of this approach?
- How do deep neural networks address this limitation?
- Write down the general formula for a deep feedforward network with L layers, expressed as a composition of functions $f_1, \dots, f_L$. Then write the formula describing how a single layer's output $\mathbf{z}_l$ is computed from the previous layer's output $\mathbf{z}_{l-1}$, in terms of weights $\mathbf{W}_l$, bias $\mathbf{b}_l$, and activation function φ_l .

Question 2

The XOR function cannot be implemented by a single perceptron (a thresholded linear combination of the inputs).

- Why not – what property of the XOR classes is the problem?
- Briefly explain how stacking two hidden units, one computing AND and one computing OR, lets a multilayer perceptron solve XOR.
- What general lesson does this example teach about the role of hidden units in a network?

Question 3

Suppose someone proposes building a deep network using only **linear** activation functions, to keep things simple.

- Explain why this would be a bad idea.
- Write down the sigmoid activation function $\sigma(a)$ and its derivative $\sigma'(a)$. For what values of $\sigma(a)$ does this derivative become (approximately) zero, and why does this matter for training deep networks?
- Write down the ReLU function and its derivative. Why does this avoid the vanishing-gradient problem that affects the sigmoid?
- Is ReLU completely free of issues? If not, name one problem it has.

Question 4

Backpropagation can be seen as a special case of automatic differentiation.

- What quantity does backpropagation compute, and why do we need it to train a network?

- b) Computing the full Jacobian of a composition of functions can be done by *forward-mode* or *reverse-mode* differentiation. Explain the difference, and state which one is used for training a neural network, and why.
- c) Suppose $\mathbf{o} = \mathbf{f}_4(\mathbf{f}_3(\mathbf{f}_2(\mathbf{f}_1(\mathbf{x}))))$. Write down the chain-rule formula for the Jacobian $\partial \mathbf{o} / \partial \mathbf{x}$ in terms of the Jacobians of $\mathbf{f}_1, \dots, \mathbf{f}_4$, making clear the order in which they are multiplied.
- d) For the cross-entropy loss layer, $z = -\sum_c y_c \log(\text{softmax}(\mathbf{x})_c)$, the Jacobian with respect to the logits $\mathbf{x}$ has a remarkably simple closed form. Write it down, using $\mathbf{p} = \text{softmax}(\mathbf{x})$.

Question 5

In a deep network, the gradient of the loss with respect to the hidden units at layer l can be written $\mathbf{g}_l = \mathbf{J}_l \mathbf{g}_{l+1}$, where $\mathbf{J}_l$ is a Jacobian.

- a) Explain informally why, after passing through many layers, this gradient can either vanish or explode, in relation to the eigenvalues of $\mathbf{J}$.
- b) Name one technique that fixes exploding gradients, and explain how it works.
- c) Name two complementary techniques used to address the (harder) problem of vanishing gradients.

Question 6

Good initial values for the weights matter a lot when the training objective is non-convex.

- a) Consider a single linear unit $o_i = \sum_{j=1}^{n_{\text{in}}} w_{ij} x_j$, with $w_{ij} \sim \mathcal{N}(0, \sigma^2)$ and inputs with $\mathbb{E}[x_j] = 0$, $\mathbb{V}[x_j] = \gamma^2$, independent of the weights. Write down the formula for $\mathbb{V}[o_i]$ in terms of n_{in} (the fan-in), σ^2 , and γ^2 , and explain why we want $n_{\text{in}} \sigma^2 \approx 1$.
- b) Write down the Xavier/Glorot formula for σ^2 , and explain what it tries to balance.
- c) Which initialization rule is typically paired with ReLU, and which with SELU?

Question 7

Large DNNs can easily overfit, and they can also be slow to train.

- a) Name three different regularization strategies used to control overfitting in deep networks, and briefly describe one of them.
- b) What is the difference between data parallelism and model parallelism when training across multiple machines?
- c) What is the difference between synchronous and asynchronous (“Hogwild”) data-parallel training?